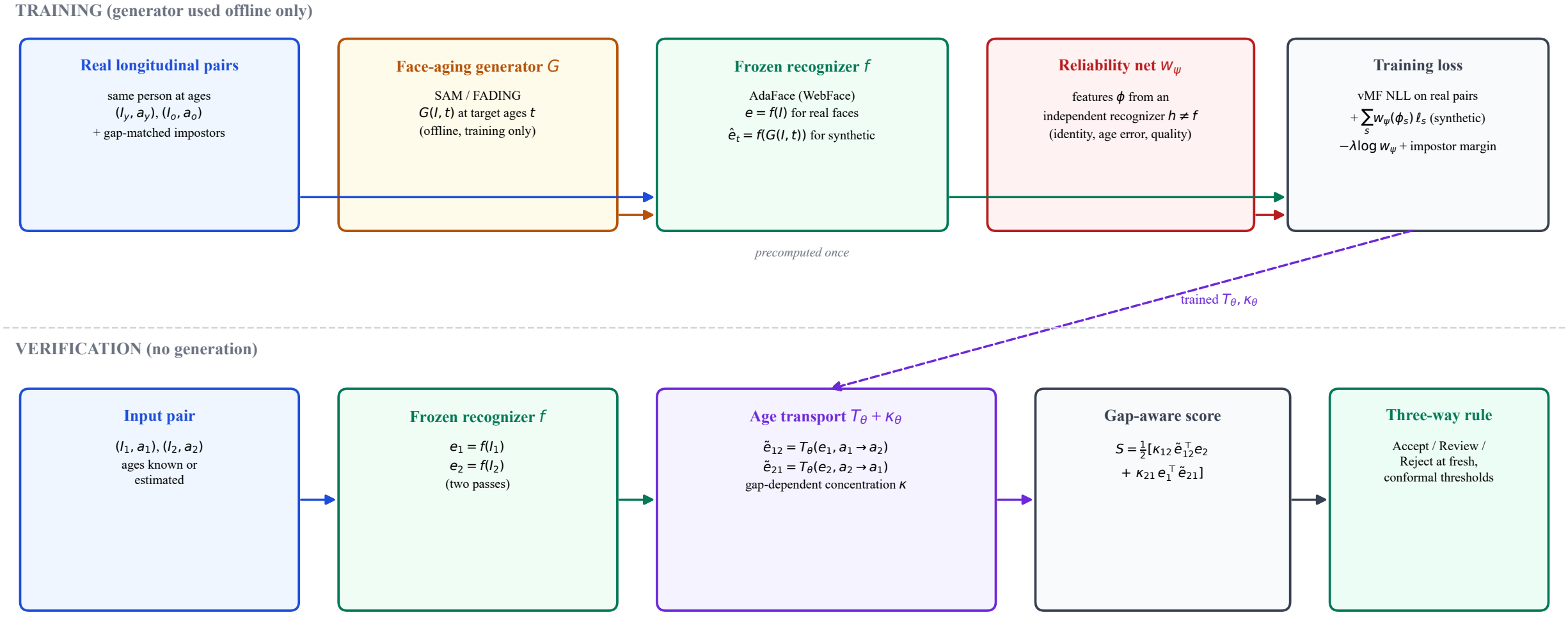

TRAINING (generator used offline only)
Real longitudinal pairs
same person at ages
(I_y, a_y), (I_o, a_o)
+ gap-matched impostors
Face-aging generator G
SAM / FADING
$G(I, t)$ at target ages t
(offline, training only)
Frozen recognizer f
AdaFace (WebFace)
$e = f(I)$ for real faces
$\hat{e}_t = f(G(I, t))$ for synthetic
Reliability net w_ψ
features ϕ from an
independent recognizer $h \neq f$
(identity, age error, quality)
Training loss
vMF NLL on real pairs
$+ \sum_s w_\psi(\phi_s)\, \ell_s$ (synthetic)
$-\lambda \log w_\psi$ + impostor margin
precomputed once
trained T_θ, κ_θ
VERIFICATION (no generation)
Input pair
(I_1, a_1), (I_2, a_2)
ages known or
estimated
Frozen recognizer f
$e_1 = f(I_1)$
$e_2 = f(I_2)$
(two passes)
Age transport $T_\theta + \kappa_\theta$
$\tilde{e}_{12} = T_\theta(e_1, a_1 \to a_2)$
$\tilde{e}_{21} = T_\theta(e_2, a_2 \to a_1)$
gap-dependent concentration κ
Gap-aware score
$S = \frac{1}{2}[\kappa_{12}\, \tilde{e}_{12}^\top e_2$
$+ \kappa_{21}\, e_1^\top \tilde{e}_{21}]$
Three-way rule
Accept / Review /
Reject at fresh,
conformal thresholds